

PHU LIEN, Viet Nam

WMO: 488260

Lat: 20.795N

Lon: 106.614E

Elev: 381

StdP: 14.49

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.7	51.7	40.7	37.9	56.3	43.6	42.5	56.4	11.9	62.9	10.7	61.5	4.1	50	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.7	93.8	84.0	91.9	83.7	90.4	83.1	86.0	91.1	84.9	89.9	83.9	88.8	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.6	184.5	89.8	83.6	178.3	88.5	82.6	172.6	87.2	51.2	91.3	49.8	90.2	48.6	89.0	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	11.5	9.8	46.1	98.0	4.9	2.8	42.6	100.0	39.7	101.7	36.9	103.2	33.4	105.3		
(5)			44.0	87.1	2.6	1.5	42.1	88.2	40.6	89.0	39.2	89.9	37.3	91.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.1	62.5	64.4	68.4	75.2	81.0	84.3	83.9	83.0	81.5	78.5	72.6	65.7
	DBStd	9.02	6.12	6.46	5.32	4.64	3.91	3.15	3.12	3.07	3.07	3.76	5.09	5.47
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	304	123	84	29	2	0	0	0	0	0	0	8	58
	CDD50	9177	387	404	571	757	961	1030	1051	1022	945	884	679	486
	CDD65	4004	44	68	135	309	496	580	586	557	495	419	236	79
	CDH74	36543	38	91	236	1690	4836	6989	6965	6188	4988	3256	1106	160
	CDH80	12451	0	5	15	256	1545	2949	2814	2245	1608	841	167	6
(14) Wind	WSAvg	4.6	3.8	3.8	3.9	4.7	5.1	5.2	5.2	4.8	4.8	4.7	4.7	4.3
(15) Precipitation	PrecAvg	63.8	1.1	0.8	1.9	2.7	7.8	9.5	11.0	12.7	9.8	3.7	1.9	0.9
	PrecMax	86.7	6.6	2.8	6.9	7.6	18.4	17.1	23.6	24.8	15.9	24.1	8.7	3.2
	PrecMin	37.9	0.0	0.0	0.0	0.7	3.0	5.6	3.6	4.8	2.6	0.0	0.0	0.0
	PrecStd	11.0	1.5	0.6	1.5	1.9	3.7	2.8	4.6	5.6	3.5	4.8	1.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.6	79.8	81.8	88.7	95.4	96.3	95.8	94.2	91.8	89.6	86.2	80.8
		MCWB	72.0	74.4	76.2	82.2	84.3	83.6	84.2	84.6	83.5	80.5	78.2	75.0
	2%	DB	74.7	76.7	78.9	85.2	91.4	93.8	92.9	92.1	90.4	87.9	83.8	78.0
		MCWB	69.7	72.2	74.6	79.7	83.6	83.5	84.0	84.0	82.8	79.3	75.9	72.6
	5%	DB	72.5	74.5	76.4	83.0	88.8	91.5	91.1	90.3	88.9	86.4	81.9	75.5
		MCWB	68.3	70.8	73.3	78.4	82.2	83.6	83.6	83.3	81.9	78.2	74.7	70.4
	10%	DB	70.5	72.5	74.6	81.0	86.8	89.6	89.3	88.5	87.3	84.8	79.9	73.3
		MCWB	67.2	70.1	72.5	77.3	80.9	82.9	82.8	82.5	80.8	77.2	73.6	68.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	75.6	77.5	82.8	86.3	87.2	87.2	87.0	85.8	83.0	80.5	76.8
		MCDB	76.1	78.9	80.1	87.3	92.0	92.0	92.1	91.8	90.4	87.4	84.5	79.5
	2%	WB	71.3	73.3	75.5	80.7	84.4	85.5	85.5	84.1	81.3	78.2	74.4	72.2
		MCDB	73.7	75.4	77.6	84.4	90.0	90.5	90.7	90.1	88.9	86.1	81.5	76.8
	5%	WB	69.6	71.9	74.2	79.1	82.9	84.3	84.3	84.0	82.8	79.8	76.9	72.2
		MCDB	71.5	73.6	75.8	82.0	87.6	89.3	89.5	88.7	87.2	84.4	80.0	74.3
	10%	WB	68.0	70.5	73.0	77.9	81.7	83.3	83.2	82.9	81.7	78.6	75.6	70.1
		MCDB	69.9	72.1	74.4	80.2	85.7	88.0	87.9	87.2	85.8	82.7	78.4	72.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.1	7.3	8.2	9.9	9.7	9.2	9.4	10.2	11.1	11.2	10.3
		MCDBR	11.9	10.6	9.9	10.7	12.5	11.8	11.8	12.0	12.3	12.9	13.1	12.4
		MCWBR	7.4	6.4	6.4	6.3	6.5	5.5	5.6	6.0	6.2	6.2	6.8	7.3
	5% WB	MCDBR	10.5	8.8	8.2	9.6	11.6	10.7	11.2	11.3	11.3	11.6	11.2	10.4
		MCWBR	7.3	6.1	6.0	6.2	6.7	5.9	6.3	6.6	6.6	6.5	6.9	7.4
(40) Clear-Sky Solar Irradiance	taub		0.589	0.625	0.789	0.743	0.547	0.509	0.483	0.536	0.584	0.681	0.584	0.612
	taud		1.772	1.737	1.476	1.592	2.083	2.210	2.277	2.099	1.933	1.680	1.840	1.750
	Ebn at Noon		216	218	190	203	245	252	259	247	233	202	216	204
	Edh at Noon		66	72	96	87	53	46	43	52	60	75	61	65
(44) All-Sky Solar Radiation	RadAvg		727	830	907	1203	1538	1599	1569	1453	1398	1262	1064	827
	RadStd		177	140	116	129	132	134	163	112	97	79	136	134

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.74	+0.63	+0.72	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page